Seth Balkrishan Memorial

BHILAIINSTITUTEOFTECHNOLOGY, DURG

(An Autonomous Institute affiliated to CSVTU, Bhilai)

SCHEME OF TEACHING AND EXAMINATION (Effective from 2020-2021 Batch)

B.Tech. (Information Technology) Seventh Semester

Sl. No.	Board of Studies (BOS)	Courses (Subject)	Course Code	Period per Week			Scheme of Examination			Total Marks	Credit
							Theory/Lab				
				L	T	P	ESE	CT	TA		
1.	Information Technology	Cryptography and Network Security	106701IT	2	1	-	100	20	30	150	3
2.	Information Technology	Cloud Computing	106702IT	2	1	-	100	20	30	150	3
3.	Management	Principles of Management	100703MG	3	-	-	100	20	30	150	3
4.	Information Technology	Professional Elective-2	Refer Table-1	3	-	-	100	20	30	150	3
5.	Refer Table-2	Open Elective-2	Refer Table-2	3	-	-	100	20	30	150	3
6.	Information Technology	Cryptography and Network Security Lab	106791IT	-	-	2	25	-	25	50	1
7.	Information Technology	Cloud Computing Lab	106792IT	-	-	2	25	-	25	50	1
8.	Information Technology	Capstone Project Phase - I	106793IT	-	-	4	50	-	50	100	2
9.	Information Technology	Internship Assessment (Report & Seminar)	106794IT	-	-	2	-	-	25	25	1
10.	Humanities	Universal Human Values and Professional Ethics	100795HM	-	-	2	-	-	25	25	-
Total				**13**	**2**	**12**	**600**	**100**	**300**	**1000**	**20**

Table 1 : Professional Elective -2

S. No.	Board of Study	Courses (Subject)	Course Code
1.	Information Technology	Soft Computing Techniques	106721IT
2.	Information Technology	E-Commerce & Strategic IT	106722IT
3.	Information Technology	Information Theory & Coding	106723IT
4.	Information Technology	Computer Vision	106724IT
5.	Information Technology	Bioinformatics	106725IT

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Table 2 : Open Elective -2

S. No.	Board of Study	Courses (Subjects)	Course Code	Eligible Branches	Link
1.	Civil Engineering	Environmental Management and Legislations	100741CE	Open to all branches	https://archive.nptel.ac.in/courses/120/108/120108004/
2.	Computer Science & Engineering	Cyber Security	100742CS	Open to all branches except CSE and IT	https://nptel.ac.in/courses/106106129/
3.	Electrical & Electronics Engineering	Power Plant Engineering	100743TE	Open to all branches	https://archive.nptel.ac.in/courses/112/107/112107291/
4.	Electrical Engineering	Solar, Wind and Biomass Energy Systems	100744EE	Open to all branches	https://archive.nptel.ac.in/courses/103/103/103103206/
5.	Electronics & Telecommunication Engineering	Artificial Intelligence and Expert Systems	100745ET	Open to all branches	https://onlinecourses.nptel.ac.in/noc19_me71/preview
6.	Information Technology	Database Management System	100746IT	Open to all branches except CSE and IT	https://nptel.ac.in/courses/106105175
7.	Mechanical Engineering	Innovation Management & Entrepreneurship	100747ME	Open to all branches	https://onlinecourses.nptel.ac.in/noc21_mg63/
8.	Management	“Principles of Management” is included as a compulsory course for all branches.			
9.	Computer Applications	Android Application Development	100748CA	Open to all branches except CSE and IT	-

Note:

(a) Abbreviations used : L-Lecture, T- Tutorial, P- Practical, ESE- End Semester Exam, CT- Class Test, TA- Teacher’s Assessment

(b) 1/4th of total strength of students subject to minimum of 20 students is required to offer an elective in the department in a particular academic session.

(c) Choice of elective course once made for an examination cannot be changed in future examinations.

(d) The duration of end semester examination of all theory papers will be of three hours

(e) “Principle of Management”will be conducted by the management / relevant discipline as decided by the Principal

(f) “Universal Human Values and Professional Ethics” will be conducted by the humanities / relevant discipline as decided by the Principal.

SYLLABUS

B.TECH. (INFORMATION TECHNOLOGY)

SEVENTH SEMESTER

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SYLLABUS

B. Tech. (Information Technology) Seventh Semester

CRYPTOGRAPHY & NETWORK SECURITY					
Course Code	**106701IT**	**L = 2**	**T = 1**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
This Course focuses towards the basic principles and practices of cryptography and network security using various cryptographic methods. Understand message and web authentication security issues. It also focuses on the practical applications that have been implemented and are in use to provide network security	**On successful completion of the course, the student will be able to:** **CO1:** Implement the conventional Encryption algorithms for confidentiality and their design principles. **CO2:** Explain the symmetric key Encryption algorithms and their design principles. **CO3:** Apply the Public Key Encryption algorithms and their design principles. **CO4:** Identify the authentication techniques, digital certificates and Public key certificates with their design principles. **CO5:** Apply concept in applications of network security, system level security issues and their countermeasures.

UNIT I: Introduction: **[CO1]**

Security Concepts: Introduction, The need for security, Security approaches, Principles of security, Types of Security attacks, Security services, Security Mechanisms, A model for Network Security

Cryptography Concepts and Techniques: Introduction, plain text and cipher text, substitution techniques, transposition techniques, encryption and decryption, symmetric and asymmetric key cryptography. **[8Hrs]**

UNIT II: Symmetric Ciphers: **[CO2]**

Modular Arithmetic, Euclidian algorithm, Symmetric key Ciphers: Block Cipher principles, DES: Principles of DES, Strength of DES, AES, Blowfish, RC5, IDEA, Block cipher operation, Stream ciphers **[5Hrs]**

UNIT III: Asymmetric (Public Key) Ciphers: : **[CO3]**

Mathematics Of Asymmetric Key Cryptography: Primes – Primality Testing – Factorization – Euler‘s totient function, Fermat‘s and Euler‘s Theorem - Chinese Remainder Theorem.

Principles of public key cryptosystems: RSA algorithm, Elgamal Cryptography, Diffie-Hellman Key Exchange, Elliptic curve Arithmetic-Elliptic curve cryptography. **[7Hrs]**

UNIT IV: **Message Authentication:** **[CO4]**

Cryptographic Hash Functions: Message Authentication, Message Digest (MD5), Secure Hash Algorithm (SHA-512).

Message authentication codes: Authentication requirements, HMAC, CMAC, Digital signatures.

Key Management and Distribution: Symmetric Key Distribution Using Symmetric & Asymmetric Encryption, Distribution of Public Keys, Kerberos, X.509 Authentication Service, Public Key Infrastructure. **[8Hrs]**

UNIT V: Network Security applications: : **[CO5]**

E-Mail Security: Pretty Good Privacy, S/MIME IP Security: IP Security overview, IP Security architecture, Authentication Header, Encapsulating security payload.

Web Security: Web security considerations, SSL and TLS, Secure electronic transaction.

Case Studies on Cryptography and security: Secure Multiparty Calculation, Virtual Elections, Single

sign On, Secure Inter-branch Payment Transactions, Cross site Scripting Vulnerability. **[8Hrs]**

Text Books:

S. No.	Title	Author(s)	Publisher
1.	Cryptography and Network Security: Principles and Practice	William Stallings	Pearson Education, Prentice Hall,
2.	Cryptography and Network Security	Atul Kahate	McGraw Hill Education (India) Private Limited;.

Reference Books:

S. No.	Title	Author(s)	Publisher
1.	Applied Cryptography: Protocols & Algorithms	Schneier & Bruce	MGH International.
2.	Cryptography and Security	Dr T R Padmanabhan N Harini	Wiley India Pvt Ltd,
3.	Handbook of Applied Cryptography	Alfred J Menezes and Scott A Vanstone.	Pearson
4.	Cryptography and Network Security	Behrouz A Forouzan and Debdeep Mukhopadhyay	Pearson
5.	Network Security: The Complete Reference	Roberta Bragg, Mark Rhodes-Ousley, Keith Strassberg	Tata McGraw-Hill Education

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B. Tech. (Information Technology) Seventh Semester

CLOUD COMPUTING					
Course Code	**106702IT**	**L = 2**	**T = 1**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
Introduce concepts of cloud computing, various security issues, cloud computing models their applications and principles of virtualization technologies and cloud computing.	**On successful completion of the course, the student can**: **CO1.** Articulate the main concepts, key technologies, strengths, and limitations of cloud computing. **CO2.** Identify the architecture and infrastructure of cloud computing, including SaaS, PaaS, IaaS, public cloud, private cloud, hybrid cloud, etc**.** **CO3.** Elucidate the core issues of cloud computing such as security, privacy, and interoperability. **CO4.** Provide the appropriate cloud computing solutions and recommendations according to the applications used. **CO5.** Implement Hardware/Software verification concept & smart phone activities.

UNIT I: Introduction To Cloud Computing: **[CO1]**

The Emergence of Cloud Computing, Cloud Based Service Offerings, Benefits of using a Cloud Model, Key Characteristics of Cloud Computing, Understanding Public & Private cloud environments, The Evolution of Cloud Computing, Hardware & Internet Software Evolution. Cloud Computing Overview Origins of Cloud computing, Cloud components – Essential characteristics, On-demand self-service, Broad network access, Comparing cloud providers with traditional IT service providers. **[7Hrs]**

UNIT II: Cloud Security Challenges & The Msp Model: **[CO2]**

Software-as-a-Service, Security Management People, Security Governance, Security Portfolio Management, Security Architecture Design, Identity Access Management (IAM), Data Security.
Evolution from the MSP Model to Cloud Computing and Software-as a-Service, The Cloud Data Centre, Basic Approach to a Data Centre-Based SOA, Open Source Software, Service- Oriented Architectures as a Step Toward Cloud Computing. **[9Hrs]**

UNIT III: Cloud Insights Architectural Influences: **[CO3]**

High-performance computing, Utility and Enterprise grid computing, Cloud scenarios, Benefits: scalability, simplicity, vendors, security, Limitations, Sensitive information - Application development- security level of third party - security benefits, Regularity issues: Government policies **[8Hrs]**

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UNIT IV: Cloud Storage Management& Building Cloud Networks: **[CO4]**

Concept of Virtualization and Load Balancing. Overview on Virtualization used for Enterprise Solutions. Key Challenges in managing Information. Identifying the problems of scale and management in big data. Designing and Implementing a Data Centre-Based Cloud Installing Open Source Cloud service. Amazon Web Services (AWS). Google Cloud Platform. **[6Hrs]**

UNIT V: Virtualization Concepts & Smartphone: **[CO5]**

Virtualization benefits, Hardware virtualization, Software Virtualization, Memory Virtualization, Storage Virtualization, Data Virtualization, Network Virtualization, Virtualization Security Recommendations,

Introduction to Various Virtualization OS VMware, KVM, Virtual Machine Security, Smartphone, Mobile Operating Systems for Smartphone's (iPhone, Windows Mobile), Google(Android) Blackberry, Ubuntu Mobile Internet. **[6Hrs]**

Text Books:

S.No.	Title	Author(s)	Publisher
1.	Cloud Computing: A Practical Approach	Toby Velte, Anthony Vote and Robert Elsenpeter	McGraw Hill

Reference Books:

S.No.	Title	Author(s)	Publisher
1.	Cloud Application Architectures : Building Applications and Infrastructures in the Cloud	George Reese	O'Reilly Media
2.	Cloud Security and Privacy: An Enterprise Perspective on Risks and Compliance	Tim Matherm, Subra Kumara swamy and Shahed Latif	O'Reilly Media

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SYLLABUS

B. Tech. (Common to all branches) Seventh Semester

PRINCIPLES OF MANAGEMENT					
Course Code	**100703MG**	**L = 3**	**T = 0**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
. This course intends to familiarize with basic management concepts, principles and practices and business ethical values.	**On successful completion of the course, the student will be able to:** **CO1:** Explain the fundamental principles of management. **CO2:** Apply planning techniques to set organizational goals and objectives. **CO3:** Demonstrate proficiency in organizing resources to achieve optimal performance. **CO4:** Apply the role of Directing, motivating and leadership in an organization. **CO5:** Acquire knowledge on control mechanisms to monitor, evaluate and manage organizational performance.

UNIT – I Management Concept **CO1**

Definition of management, science or art, manager vs entrepreneur; Types of managers-managerial roles and skills; Evolution of management- scientific, human relations, system and contingency approaches; Types of Business Organizations, sole proprietorship, partnership, company, public and private enterprises; Organization culture and environment; Current trends and issues in management. **[8 Hrs]**

UNIT – II Planning **CO2**

Nature and purpose of Planning, types of Planning, objectives, setting objectives, policies, Strategic Management, Planning Tools and Techniques, Decision making steps & processes**.** **[7 Hrs]**

UNIT – III Organizing **CO3**

Nature and purpose of Organizing, formal and informal organization, organization structure , types line and staff authority, departmentalization, delegation of authority, centralization and decentralization, job design, human resource management, HR planning, Recruitment selection, Training & Development, Performance Management, Career planning and Management. **[7 Hrs]**

UNIT – IV Directing Motivation and Leadership **CO4**

Directing, individual and group behavior, motivation, motivation theories, motivational techniques, job satisfaction, job enrichment leadership, types & theories of leadership, effective communication. . **[7 Hrs]**

		July 2023	1.00	Applicable for
Chairman (AC)	Chairman (BoS)	Date of Release	Version	AY 2023-24 Onwards

UNIT – V Controlling **CO5**

Controlling system and process of controlling, budgetary and non-budgetary control techniques, use of computers and IT in management control, productivity problems and management, control and performance, direct and preventive control, reporting. **[7 Hrs]**

Text Books:

S. No.	Title	Author(s)	Publisher
1.	Principles of Management	Koontz, Weihrich, Aryasri	Mc Graw Hill
2.	Principles of Management	Tripathy PC & Reddy PN	Tata Mc Graw Hill

Reference Books:

S. No.	Title	Author(s)	Publisher
1.	The Practice of Management	Peter F. Drucker	Allied Publishers
2.	Principles of Management	T. Ramaswamy	Himalaya Publishing House
3.	Management	Robins S.P. and Couiter M.	Prentice Hall India

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SYLLABUS

B. Tech. (Information Technology) Seventh Semester

SOFT COMPUTING TECHNIQUES					
Course Code	**106721IT**	**L = 3**	**T = 0**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
The objective of this course is to familiarize with soft computing concepts, use the idea of Neural networks, fuzzy logic and use of heuristics based on human experience, apply the concepts of Genetic algorithm in real-life applications and introduce to hybrids of soft computing techniques.	**On successful completion of the course, the student will be able to**: **CO1.** Describe soft computing techniques and its applications. **CO2.** Develop the artificial neural networks and its applications. **CO3.** Develop the fuzzy logic sets and membership function and defuzzification techniques. **CO4.** Analyze the single-objective optimization problems using GAs. **CO5.** Justify on hybrid control schemes according to .problem statements.

UNIT I: Introduction of Soft computing **[CO1]**

Concept of computing systems, What is Soft Computing? Difference between Hard and Soft computing, Requirement of Soft computing, Characteristics of Soft Computing, Introduction to intelligent systems- Soft computing techniques, Applications of Soft Computing **[7Hrs]**

UNIT II: Artificial Neural Network **[CO2]**

Artificial Neural Networks - biological neurons and its working, Basic models of artificial neural networks – Connections, Learning, Activation Functions, Single layer Perceptron, Back Propagation networks, Architecture of Backpropagation(BP) Networks, Backpropagation Learning, Variation of Standard Back propagation Neural Network, Recurrent Neural Network (RNN), Adaptive Resonance Theory (ART) based network, Radial basis function network, online learning algorithms, Back Propagation, RTRL algorithms– Reinforcement learning. Different ANNs architectures, Training techniques for ANNs, Modelling of non-linear systems using ANN, Applications of ANNs to solve some real-life problems. **[9Hrs]**

UNIT III: Fuzzy logic **[CO3]**

Fuzzy logic: Introduction to Fuzzy logic, Fuzzy sets and membership functions, Operations on Fuzzy sets, Fuzzy relations- operations on fuzzy relations, rules, propositions, implications and inferences, Fuzzy membership functions, fuzzification, Methods of membership value assignments, Modelling of non-linear systems using fuzzy models. Defuzzification techniques, Some applications of Fuzzy **[8 Hrs]**

UNIT IV: Genetic Algorithms **[CO4]**

Evolutionary programs – Genetic algorithms, genetic programming and evolutionary programming - Genetic Algorithm versus Conventional Optimization Techniques, Basic GA framework and different GA architectures, Genetic operators- different types of crossover and mutation operators - Optimization problems using GA-discrete and continuous - Single objective and multi-objective problems - Procedures in evolutionary programming. **[6Hrs]**

UNIT V: Hybrid Systems **[CO5]**

Fuzzification and rule base using ANN – Neuro fuzzy systems, Fuzzy Neural systems, Genetic Fuzzy systems- Optimization of membership function and rule base using Genetic Algorithm , Neuro-Genetic Hybrid Systems,. **[6Hrs]**

Text Books:

S.No.	Title	Author(s)	Publisher
1.	Principles of Soft Computing	S.N. Sivanandam & S.N. Deepa	Wiley India Pvt. Ltd.
2.	Fundamentals of Neural Networks	Laurence Fausett	Prentice Hill

Reference Books:

S.No.	Title	Author(s)	Publisher
1.	Fuzzy Logic with Engineering Applications	Timothy J. Ross	McGraw-Hill
2.	Genetic Algorithms: Search, Optimization and Machine Learning	Davis E. Goldberg	Addison Wesley
3.	Neural Networks, Fuzzy Logic and Genetic Algorithms	S. Rajasekaran and G.A.V. Pai	PHI

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SYLLABUS

B. Tech. (Information Technology) Seventh Semester

E-COMMERCE AND STRATEGIC IT					
Course Code	**106722IT**	**L = 3**	**T = 0**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
This course provides an introduction to information systems for business and management. It is designed to familiarize students with organizational and managerial foundations of systems, the technical foundation for understanding information systems	**On successful completion of the course, the student will be able to**: **CO1.**Explain the basic concepts and technologies used in the field of management information systems; **CO2.** Select different types of management information systems; **CO3.** Describe the processes of developing and implementing information systems; **CO4.** Justify the security issues of information systems; **CO5.** Describe the ethical, social issues of information systems;

UNIT V: Hybrid Systems **[CO1]**

E-commerce and its Technological Aspects Overview of developments in Information Technology and Defining E-Commerce: The scope of E commerce, Electronic Market, Electronic Data Interchange, Internet Commerce, Benefits and limitations of E-Commerce, Produce a generic framework for E-Commerce, Architectural framework of Electronic Commerce, Web based E Commerce Architecture **[7Hrs]**

UNIT II: Consumer Oriented E Commerce E-Retailing: **[CO2]**

Traditional retailing and e retailing, Benefits of e retailing, Key success factors, Models of e retailing, Features of e retailing. E services: Categories of e-services, Web-enabled services, matchmaking services, Information-selling on the web, e entertainment, Auctions and other specialized services. Business to Business Electronic Commerce **[9Hrs]**

UNIT III: Electronic Data Interchange: **[CO3]**

Benefits of EDI, EDI technology, EDI standards, EDI communications, EDI Implementation, EDI Agreements, EDI Security. Electronic Payment Systems, Need of Electronic Payment System: Study and examine the use of Electronic Payment system and the protocols used, Study Electronic Fund Transfer and secure electronic transaction protocol for credit card payment. Digital economy: Identify the methods of payments on the net – Electronic Cash, cheques and credit cards on the Internet.**[8 Hrs]**

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UNIT IV: Security in E Commerce Threats in Computer Systems: **[CO4]**

Virus, Cyber Crime Network Security: Encryption, Protecting Web server with a Firewall, Firewall and the Security Policy, Network Firewalls and Application Firewalls, Proxy Server. **[6 Hrs]**

UNIT V: Issues in E Commerce Understanding Ethical, Social and Political issues in E-Commerce: **[CO5]**

A model for Organizing the issues, Basic Ethical Concepts, Analyzing Ethical Dilemmas, Candidate Ethical Principles Privacy and Information Rights: Information collected at E-Commerce Websites, The

Concept of Privacy, Legal protections Intellectual Property Rights: Types of Intellectual Property protection, Governance. **[6 Hrs]**

Text Books:

S.No.	Title	Author(s)	Publisher
1.	" Electronic Commerce",	Elias. M. Awad	Prentice-Hall of India Pvt Ltd.
2.	Electronic Commerce-A Manager's guide", Addison-Wesley.	Ravi Kalakota, Andrew B. Whinston	Prentice-Hall of India Pvt Ltd

Reference Books:

S.No.	Title	Author(s)	Publisher
1.	Electronic Commerce–A Managerial Perspective",	Efraim Turban, Jae Lee, David King, H.Michael Chung. Elias M Award	Addison-Wesley.
2.	Electronic Commerce from Vision to Fulfilment	Judy Strauss, Adel El-Ansary, Raymond Frost	Pearson Education.

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B. Tech. (Information Technology) Seventh Semester

Information Theory and Coding					
Course Code	**106723IT**	**L = 3**	**T = 0**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
The main Objective to include this subject in Information Technology discipline is to design and study of the mathematical model of communication systems and their corresponding probabilistic predictions.	**On successful completion of the course, the Students will be able to:** **CO1.**Acknowledge about the different probability of erroneous conditions in communication and how to control it. **CO2.**Optimize the channel bandwidth and different conditions to improve efficiency of the communication system. **CO3.** Demonstrate the basic concept of Source Coding Techniques to enhance the performance of Communication Systems. **CO4.** Compare different types of channels. Understand basic concepts of complexity of cryptographic security methods and their practical applications. **CO5.** Apply and control specific coding methods and be able to calculate the rate and error probabilities achieved.

UNIT I: Information Theory **[CO1]**

Uncertainty, Information, Information rate, mutual information; Marginal, conditional and joint Entropies; Shannon's concept of information; Shannon's measure of information; Model for Communication system; Source coding and line/channel coding; channel mutual information capacity (Bandwidth) systems. **[7Hrs]**

UNIT II: Source Coding **[CO2]**

Model for source coding theorem; entropy of binary memory less source and its extension to discrete memory less source, Source with memory and its entropy, Encoding of the source output, source coding theorem, Huffman coding, Shannon-Fano coding. **[9Hrs]**

UNIT III: Channel Coding **[CO3]**

Binary Symmetric Channel, mutual information & its properties, Channel capacity, channel coding theorem and its application to BSC, Shannon's theorem on channel capacity, capacity of a channel of infinite bandwidth, bandwidth - S/N trade off, practical communication systems in light of Shannon's theorem. **[8Hrs]**

UNIT IV: Error Control Coding **[CO4]**

BLOCK CODES: Types of codes; Error detecting codes, Parity check codes- Single parity codes; Parity check polynomials; error correcting codes, Block codes; Linear block codes, Cyclic codes - Syndrome calculation, Encoder and decoder – CRC, Hamming codes, Hamming weight, Hamming distance, Minimum distance decoding, Hamming codes. **[6Hrs]**

UNIT V: Error Control Coding **[CO5]**

CONVOLUTIONAL CODES: Convolutional codes- Feedforward Convolutional Encoder, Trellis Representation, Viterbi Decoder for convolutional codes, Recursive convolutional encoders, Recursive convolutional encoders, Puncturing, Free distance of convolutional codes. LDPC codes. **[6 Hrs]**

Text Books:

S. No.	Title	Author(s)	Publisher
1.	Communication Systems	R. P. Singh and S. D. Sapre	TMH
2.	Information Theory, Coding and Cryptography	Ranjan	TMH
3.	Communication Systems –II	J.S.Chittode	TMH

Reference Books:

S. No.	Title	Author(s)	Publisher
1.	Elements of Information Theory	Thomas M. Cover, Joy A.John	Wiley & Sons
2.	Digital Communications.	John G. Proakis,	McGraw Hill

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SYLLABUS

B. Tech. (Information Technology) Seventh Semester

COMPUTER VISION					
Course Code	**106724IT**	**L = 3**	**T = 0**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
The course objective is to make the students understand about basic concept of image processing; acquire knowledge pattern analysis and finally deploy projects related to computer vision.	**On successful completion of the course, the student will be able to:** **CO1**: Examine both the theoretical and practical aspects of computing with images **CO2**: Describe the foundation of image formation, measurement, and analysis. **CO3**: Demonstrate the geometric relationships between 2D images and the 3D world. **CO4**: Calculate the concept extraction. **CO5**: List the principles of state-of-the-art deep neural networks.

UNIT I: Introduction **[CO1]**

Overview, computer imaging systems, lenses, Image formation and sensing, Image analysis: 2D and 3D object recognition, image segmentation, motion detection e.g. Single particle tracking, video tracking,optical flow, medical scan analysis,3D Pose Estimation, pre-processing and Binary image analysis **[7Hrs]**

UNIT II: Edge Detection **[CO2]**

Edge detection, Edge detection algorithms: algorithms: Sobel, Canny, Prewitt, Roberts, and fuzzy logic methods. Edge detection performance, Hough transform, corner detection and it's applications **[8Hrs]**

UNIT III: Image Segmentation and Restoration **[CO3]**

Segmentation, Different types of image segmentation: Thresholding Segmentation, Edge-Based Segmentation, Region-Based, Watershed, Clustering-Based Algorithms, Neural Networks for Segmentation, Morphological filtering, Image Restoration - degradation model, Properties, Noise models, Fourier transform. **[7Hrs]**

UNIT IV: Feature Extraction **[CO4]**

Feature extraction, shape, histogram, color, spectral, texture, using CV IP tools, Feature analysis, feature vectors, similarity measures: city block distance, Euclidean distance, and cosine distance, data preprocessing and it's steps. **[7Hrs]**

UNIT V: Pattern Analysis **[CO5]**

Clustering: K-Means, K-Medoids, Mixture of Gaussians, Classification: Discriminant Function, Supervised, Un-supervised, Semi-supervised, Classifiers: Bayes, KNN, Recent trends in Activity Recognition, computational photography, Biometrics. **[7Hrs]**

Text Books:

S.No.	Title	Author(s)	Publisher
1.	Multiple View Geometry in Computer Vision	Richard Hartley, and Andrew Zisserman	Cambridge University Press
2.	Digital Image Processing	Rafael Gonzalez, and Richard Woods	Prentice Hall

Reference Books:

S.No.	Title	Author(s)	Publisher
1.	Computer Vision: A Modern Approach	D. Forsythe and J. Ponce	Prentice-Hall
2.	Computer Vision: Algorithms and Applications	Richard Szeliski	Springer, 2010 online version

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B. Tech. (Information Technology) Seventh Semester

BIOINFORMATICS					
Course Code	**106725IT**	**L = 3**	**T = 0**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
The course objective is to make the students understand about basic structure of human cell and cell membrane; also acquire knowledge about Genomics algorithms and finally deploy projects related to bioinformatics.	**On successful completion of the course, the student will be able to:** **CO1**: Discuss and describe the cell structure. **CO2**: Solve the problem using structure and functioning of Cell membrane. **CO3**: Demonstrate the concept of Genomic. **CO4**: Classify and analyze the Genomic algorithms. **CO5**: Implement the concept of RNA and DNA

UNIT I: Introduction **[CO1]**

Cell: An introduction and classification of organism by cell structure, cytosol, compartmentalization of eukaryotic cells, Cell fractionation. Cell Membrane and Permeability: Chemical components of biological membranes, organization and fluid Mosaic Model **[7Hrs]**

UNIT II: Cell Structure **[CO2]**

Membrane vacuolar system, cytoskeleton and cell motility: Structure and function of microtubules, Microfilaments, Intermediate filaments. Endoplasmic reticulum: Structure, function including role in protein segregation. Golgi complex: Structure, biogenesis and functions including role in protein secretion Lysosomes: Vacuoles and microbodies: Structure and functions. Ribosomes: Structure and function including role in protein synthesis. **[8Hrs]**

UNIT III: Genomics **[CO3]**

Genomics: Genome Annotation, Genome Assembly, Structural and Functional Genomics. Comparative Genomics, Microarray: technique, Design, Analysis, Drug target identification. **[7Hrs]**

UNIT IV: Genetic Algorithms **[CO4]**

Artificial Neural Networks, Genetic algorithm, Bayesian modeling, Monte Carlo Simulation Method, Markov Models and their application. **[7Hrs]**

UNIT V: PERL **[CO5]**

Perl basic: Variables, Perl operations, Program to store DNA sequence, Concatenating DNA fragment, Transcription: DNA to RNA, Subroutines, scoping and subroutines, command-line arguments and arrays passing data to subroutines, modules and libraries of subroutines, fixing bugs. **[7Hrs]**

Text Books:

S.No.	Title	Author(s)	Publisher
1.	Introduction to Bioinformatics	Arthur K. Lesk	Oxford University Press
2.	Algorithms on Strings, Trees and Sequences	Dan Gusfield	Cambridge University Press

Reference Books:

S.No.	Title	Author(s)	Publisher
1.	Bioinformatics Sequence and Genome Analysis	David W. Mount	Cold Spring Harbor Laboratory Press.
2.	Beginning Perl for Bioinformatics: An introduction to Perl for Biologists	James Tindall	OReilley Media

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SYLLABUS

B. Tech. (Civil Engineering) Seventh Semester

OPEN ELECFIVE-2: ENVIRONMENTAL MANAGEMENT AND LEGISLATIONS					
(Open Elective Course : Open to all branches)					
Course Code	**100741CE**	**L = 3**	**T = 0**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
The course objective is to make the students study about various International and National Protocols. Also they will be able to gain knowledge about the important legislations and be familiar with LCA and environmental auditing.	**On successful completion of the course, the student will be able to:** **CO1:** Develop an understanding about various International and National Protocols. **CO2:** Apply the ISO procedures. **CO3:** Apply the knowledge gained for green procurement. **CO4**: Efficiently handle the environmental management instruments. **CO5**: Apply concept of sustainable development for engineering design and decision-making.

UNIT I: METHODOLOGY **CO1**

Methodology of environmental management – Review various national and international protocols. **[7 Hrs.]**

UNIT II: FUNDAMENTALS OF ENVIRONMENTAL MANAGEMENT **CO2**

Fundamentals of Environmental Management and ISO 14000 series: Environmental management Plans, principles and elements. The ISO 14001- Environmental management systems standard. **[8 Hrs.]**

UNIT III: LIFE CYCLE ANALYSIS **CO3**

Life cycle analysis, LCA steps, Framework and Methodology, Limitations of LCA, Environmental auditing **[7 Hrs.]**

UNIT IV: ENVIRONMENTAL MANAGEMENT PROGRAMMES **CO4**

Environmental Management Programmes – Economic incentive and disincentives as instruments for environmental management – Tax, subsidies, fee, and tradable permits. **[7 Hrs.]**

UNIT – V: MAJOR LEGISLATIONS AND SUSTAINABLE DEVELOPMENT **CO5**

Major legislations: Clean Water and Air Act. Principles of sustainable development, Related Issues and implications of finite biosphere and complexities for engineering design and decision-making. **[7 Hrs.]**

Text Books:

S. No.	Title	Author(s)	Publisher
1.	Environmental Engg & Management	Dhameja. S. K	S.K Kataria & Sons
2.	Environmental Law and policy in India – Cases, Material and Status	Rosencranz, S. Divan, M.L. Noble	Tripathi Pvt. Ltd.
3.	Encyclopedia of Environmental Analysis and Remediation Vol. 1-8	Meyers A. Robert	John Wiley & Sons
4.	Accident Prevention Manual for Administration & Programs (Occupational Safety and Health Series)	Philip Hagan	National Safety Council

Reference Books:

S. No.	Title	Author(s)	Publisher
1.	The theory of Environmental policy	W.J. Banmol and W.E. Dates	Cambridge University
2.	Legal aspects of environmental pollution and its management	S. Musharraf	C.B.S. Publishers
3.	Introduction to Environmental Engineering & Science	G.M. Masters	Prentice Hall
4.	Environmental Science & Engineering	J.G. Henry and G. W. Heike	Prentice Hall
5.	Environmental Quality Management	Lohani B. N. and North A. M.	South Asian

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SYLLABUS

B. Tech. (Computer Science and Engineering) Seventh Semester

CYBER SECURITY					
(Open Elective Course : Open to all branches except CSE & IT)					
Course Code	**100742CS**	**L = 3**	**T = 0**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
The course objective of Cyber Security is to provide students with the knowledge of Information system, understanding the various types of cyber threats. Students will learn about risk assessment and management, cryptography, access control and other essential aspects of cyber security. They will understand the security aspects of Social Media. They will also learn about legal and ethical issues related to cyber security.	**On successful completion of the course, the student will be able to:** **CO1:** Demonstrate basic understanding about the information system, information and cyber security. **CO2:** Discuss application and data security and Security Technology. **CO3:** Describe various types of Fraud Techniques and Security Threats. **CO4:** Demonstrate the understanding of various privacy and security concerns on online Social media its legal aspects and best practices. **CO5:** Discuss cyber-crimes and cyber laws (Security Standards and Polices).

UNIT – I **(CO1)**

Introduction to information systems: Types of information Systems, Development of Information Systems, Introduction to information security, Need for Information security, Threats to Information Systems, Information Assurance Fundamentals (Authentication, Authorization, Non-repudiation, Confidentiality, Integrity, availability), Cyber Security, and Security Risk Analysis. **[7Hrs]**

UNIT – II **(CO2)**

Security Considerations: Application security (Database, E-mail and Internet), Data Security-Backups, ArchivalStorage and Disposal of Data, Security Technology-Firewalls and VPNs, Intrusion Detection, Access Control, Basic Cryptography. **[7Hrs]**

UNIT – III **(CO3)**

Fraud Techniques: Phishing, Smishing, Vishing, and malicious mobile code, Rogue antivirus, Click fraud, Threat Infrastructure (Botnets, Fast Flux and advanced fast flux), Security Threats to E-Commerce, Electronic Payment System, e-Cash, Credit/Debit Cards.

Security Threats: Viruses, Worms, Trojan Horse, Bombs, Trapdoors, Spoofs, E-mail viruses, Macro viruses, Malicious Software, Network and Denial of Services Attack, Brute Force, and Dictionary Attacks. **[8Hrs]**

UNIT – IV **(CO4)**

Social Media Overview and Security: Types of Social networks and Social media, Social media monitoring, Social media marketing, Challenges, opportunities and pitfalls in online social network, Security issues related to social media, Flagging and reporting of inappropriate content, Laws regarding posting of inappropriate content, Best practices for the use of Social media. **[7Hrs]**

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UNIT – V **(CO5)**

Cyber Crimes and Criminals: Definition of cyber-crime, types of cyber-crimes and types of cyber-criminals.

Security Standards and Polices: Overview of IT Act 2000, Limitations of IT Act, Digital Signature, Legal Recognition of Electronic Records, Legal Recognition of Digital Signature. Introduction to IPR, Patent Law, Trademark Law, Copyright Copy Law, Software License. **[7Hrs]**

Text Books:

S. No.	Title	Author(s)	Publisher
1.	Introduction to Information Security and Cyber Law	Dr. Surya Prakash Tripathi, Ritendra Goyal, Praveen Kumar Shukla	Dreamtech Press
2.	Cyber Security Understanding Cyber Crimes, Computer Forensics and Legal Perspectives	Sumit Belapure and Nina Godbole	Wiley India

Reference Books:

S. No.	Title	Author(s)	Publisher
1.	Cyber Security Essentials	James Graham, Richard Howard and Ryan Olson	CRC Press
2.	Cyber Laws: Intellectual property & E Commerce Security	Kumar K.	Dominant Publisher
3.	Intellectual Property Rights (Patents & Cyber Law)	Dr. A. Srinivas	Oxford University Press
4.	Cyber Laws And IT Protection	Harish Chander	PHI Learning
5.	Computer Security: Art and Science	Matt Bishop	Addison Wesley

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SYLLABUS

B. Tech. (Electrical and Electronics Engineering) Seventh Semester

POWER PLANT ENGINEERING					
(Open Elective Course : Open to all branches)					
Course Code	**100743TE**	**L = 3**	**T = 0**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
The course provides theoretical knowledge of various Power Plants along with its layout, auxiliary equipment's, working, merits, demerits, selection of site, impact on environment and capital & operating cost.	**On successful completion of the course, the student will be able to:** **CO1:** Evaluate various factors related to plant economics. **CO2:** Describe concepts of steam and Coal Based Thermal Power Plants. **CO3:** Describe concepts of Nuclear Power Plants. **CO4:** Explain the concepts of Gas Turbine and Combined Cycle Power Plants. **CO5:** Evaluate tariffs and explain environmental impacts of power plants.

UNIT – I **(CO1)**

Power plant economics and selection: Power and energy, different types of power plant, Economics of power generation: load curves, maximum demand, average load diversity factor, capacity factor, utilization factor, plant operating factor, effect of plant type on costs, fixed elements, energy elements, customer elements and investor's profit; depreciation and replacement, Economics of plant selection, other considerations in plant selection. **[7Hrs]**

UNIT – II **(CO2)**

Steam power plant: Introduction, Site selection of Steam power plants, Merits and Demerits of steam power plant, General layout of steam power plant, Steam Power Plant Auxiliaries.

Coal Based Thermal Power Plants: Classification of coal, selection of coal for power plants, Layout of modern coal power plant, FBC boilers, fuel and ash handling, draught system, feed water treatment, binary cycles and cogeneration systems. **[7Hrs]**

UNIT – III **(CO3)**

Nuclear Power Plants: Introduction, Site selection of nuclear power plants, Constituents of Nuclear power plant and Layout, nuclear reactor-Main parts and their functions, Boiling Water Reactor (BWR), Pressurized Water Reactor (PWR), CANDU Reactor, Pressurized Heavy Water Reactor (PHWR), Fast Breeder Reactors (FBR), gas cooled and liquid metal cooled reactors, safety measures for nuclear power plants. **[7Hrs]**

UNIT – IV **(CO4)**

Diesel power plant: Introduction, Selection of site for diesel electric power plant, General layout, Components of Diesel power plant, merits, demerits and application.

Gas turbine power plant: Introduction, Selection of site for gas turbine power plant, Layout of gas turbine power plant, Elements of gas turbine power plants, operation and maintenance, combined steam and gas turbine power plants. **[7Hrs]**

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UNIT – V **(CO5)**

Energy and Environmental impacts: Energy, Economic and Environmental Issues of Power Plants**,** Pollution due to power generation, pollution control technologies including waste disposal options for coal and nuclear plants, impact of low power factor and remedial measures.

Power tariffs: Types of tariffs, Types of consumer**.** **[7Hrs]**

Text Books:

S. No.	Title	Author(s)	Publisher
1.	Power Plant Engineering	P.K. Nag	Tata McGraw Hill.
2.	Power Plant Engineering	F.T. Morse	Affiliated East-West Press
3.	Power Plant Technology	El-Vakil	Tata McGraw Hill.

Reference Books:

S. No.	Title	Author(s)	Publisher
1.	Power Plant Engineering	Gupta B. R.	Eurasia publications
2.	A course in Power Plant Engineering	Arora and Domkundwar	Dhanpat Rai
3.	A course in Power System Engineering	J. B. Gupta	S K Kataria & Sons

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SYLLABUS

B. Tech. (Electrical Engineering) Seventh Semester

SOLAR, WIND AND BIOMASS ENERGY SYSTEMS					
(Open Elective Course : Open to all branches)					
Course Code	**100744EE**	**L = 3**	**T = 0**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
• To introduce the renewable energy conversion technologies. • To familiarize the properties of solarwind and biomass energy systems. • To analyze the feasibility of power production from solar, wind and biomass sources.	**At the end of this course, students will demonstrate the ability to** **CO1.** Discuss the energy scenario and the consequent growth of the power generation from renewable energy sources. **CO2.** Explain the basic physics of solar power generation. **CO3.** Discuss the application of solar generation. **CO4.** Discuss the issues related to wind turbines. **CO5.** Explain the working principle and applications of biomass energy generation.

Unit 1 **(CO1)**

Introduction- Basic Concept of Energy, Source of Solar Energy, Formation of the Atmosphere, Solar Spectrum. Solar Constant, Air Mass, Sun Earth Angles Solar Radiation, Pyrheliometer, Pyrometer, Sunshine Recorder, Solar Radiation on a Horizontal Surface, Extra-terrestrial Region, Terrestrial Region and on an Inclined surface, Total Solar Radiation on any Surface. **[8Hrs]**

Unit 2 **(CO2)**

Solar Thermal system- Principle of Conversion of Solar Radiation into Heat, Solar thermal collectors – General description and characteristics, Flat plate collectors, Heat transfer processes –Solar concentrators (parabolic trough, parabolic dish, Central Tower Collector), Solar heating system, Air conditioning and Refrigeration system, Pumping system, solar cooker, Solar Furnace. **[8Hrs]**

Unit 3 **(CO3)**

Solar PV Systems- Fundamentals of Solar Cells, Photovoltaic Effect, Basic Parameters of the Solar Cell. Photovoltaic (PV) Module and Array. Mono and Poly Crystal, Thin-Film, Single Junction and Multi-Junction Modules. Packing Factor, Efficiency of the PV Module, Series and Parallel Combination of PV Modules. Maximum Power Point Tracker (MPPT). **[8Hrs]**

Unit 4 **(CO4)**

Wind Turbines - Introduction, Origin of Winds, Nature of Winds, Classification of Wind Turbines - Wind Turbine Aerodynamics - Basic principles of wind energy extraction, Extraction of wind turbine power, Weibull Distribution, Wind power generation Curve, Betz's Law. **[6Hrs]**

Unit 5 **(CO5)**

Biomass as energy source- Biomass conversion processes, Application of biomass conversion products, Biomass properties for conversion process, Physical properties, Thermal properties, Biomass pretreatment processes, Biodiesel and bioethanol. **[6Hrs]**

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Text Books:

S. No.	Title	Author(s)	Publisher
1.	Wind Power Plants and Project Development	Earnest J. and T. Wizelius	PHI
2.	Solar Energy: Principles of Thermal Collection and Storage	S. P. Sukhatme	McGraw Hill
3.	Renewable Energy Applications	G. N. Tiwari and M. K. Ghosal	Narosa

Reference Books:

S. No.	Title	Author(s)	Publisher
1.	Solar Energy Engineering,	A.A.M. Saigh (Ed)	Academic Press
2.	Wind Energy Systems and Applications,	D. P. Kothari, S. Umashankar	Narosa
3.	Non-Conventional Energy Resources,	Khan B. H.	Tata McGraw Hill

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SYLLABUS

B. Tech. (Electronics and Telecommunication Engineering) Seventh Semester

ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS					
(Open Elective Course : Open to all branches)					
Course Code	**100745ET**	**L = 3**	**T = 0**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
The objective of the course is to make the student familiar with the basics components of Artificial Intelligence and Expert System.	***On successful completion of the course, the student will be able to:*** **CO1:** Describe different models of intelligent agents and search methods. **CO2:** Use different paradigms about representing knowledge. **CO3:** Classify between reasoning and decision making in uncertain world. **CO4:** Analyze different modules of construct plans and methods for generating knowledge. **CO5:** Apply the concepts of Expert Systems.

UNIT – I **(CO1)**

Introduction: Introduction to AI: Intelligent Agents, Perception, Natural Language Processing, Problem, Solving Agents, Searching For Solutions: Uniformed Search Strategies, Informed Search Strategies.

. **[8Hrs]**

UNIT – II **(CO2)**

Knowledge and Reasoning: Adversarial search, Optimal and imperfect decisions, Alpha, Beta pruning, Logical agents: Propositional logic, First order logic, Syntax and semantics, Using first order logic, Inference in first order logic. **[7Hrs]**

UNIT – III **(CO3)**

Uncertain Knowledge and Reasoning: Uncertainty, Acting under uncertainty, Basic probability notation, Axioms of probability, Baye's rule, Probabilistic reasoning, Making simple decisions. **[7Hrs]**

UNIT – IV **(CO4)**

Planning and Learning:

Planning: Planning Problem, Partial Order Planning, Planning and Acting In Non-Deterministic Domains – **Learning:** Learning Decision Trees, Knowledge in Learning, Neural Networks, Reinforcement Learning – Passive And Active. **[7Hrs]**

UNIT – V **(CO5)**

Expert Systems:

Definition – Features of an Expert System, Organization, Characteristics, Prospector, Knowledge Representation in Expert Systems, Expert system tools, MYCIN, EMYCIN. **[7Hrs]**

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Text Books:

S. No.	Title	Author(s)	Publisher
1.	Artificial Intelligence: A Modern Approach	Stuart Russel and Peter Norvig	Pearson Education.
2.	A Guide to Expert Systems	Donald A.Waterman	Pearson Education

Reference Books:

S. No.	Title	Author(s)	Publisher
1.	Artificial Intelligence – Structures and Strategies for Complex Problem Solving	George F.Luger	Pearson Education
2.	Artificial Intelligence	Elain Rich and Kevin Knight	Tata McGraw Hill
3.	Foundations of Artificial Intelligence and Expert Systems	Janakiraman, K.Sarukesi	Macmillan Series in Computer Science
4.	Introduction to Artificial Intelligence and Expert Systems	W. Patterson	Prentice Hall of India

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SYLLABUS

B.Tech. (Seventh Semester)

DATABASE MANAGEMENT SYSTEM (DBMS)					
(Open Elective Course : Open to all branches except CSE & IT)					
Course Code	**100746IT**	**L = 3**	**T = 0**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
To Teach the basic database concepts, applications, data models, schemas and instances and entity Relationship model for a database. To Demonstrate the use of constraints and relational algebra operations and describe the basics of SQL and construct queries using SQL. To Emphasize the importance of normalization in databases. To Demonstrate the basic concepts of Query and transaction Processing and also familiarize the concepts of transaction processing and concurrency control.	**On successful completion of the course, the student will be able to:** **CO1:** Use the basic concepts of Database Systems in Database design and Design a Database using ER Modelling. **CO2:** Apply SQL queries to interact with Database. **CO3:** Explain fundamental operations of relational algebra and use various Pl/SQL concepts. **CO4:** Apply normalization on database design to eliminate anomalies. **CO5:** Analyze database transactions and can control them by applying ACID properties, concurrency control techniques and recovery procedure.

UNIT I: **(CO1)**

Introductory Concepts of DBMS: Introduction and application of DBMS, Data Independence, Database System Architecture – levels, Mapping, Database users and DBA, Entity-Relationship model, constraints, keys, Design issues, E-R Diagram, Extended E-R features- Generalization, Specialization, Aggregation. **[8 Hrs]**

UNIT II: **(CO2)**

BASIC SQL: Simple Database schema, data types, table definitions (create, alter), different DML operations (insert, delete, update), basic SQL querying (select and project) using where clause, arithmetic & logical operations, SQL functions (Date and Time, Numeric, String conversion). **[7 Hrs]**

UNIT III: **(CO3)**

Relational Model: Translating E-R model into Relational mode, Relational Algebra, Fundamental operations, DCL PL/SQL Concepts, Cursors, Stored Procedures, Stored Functions, Database Integrity – Triggers. **[7 Hrs]**

UNIT IV: **(CO4)**

Schema Refinement (Normalization): Purpose of Normalization or schema refinement, concept of functional dependency, normal forms based on functional dependency(1NF, 2NF and 3 NF), concept of surrogate key, Boyce-codd normal form(BCNF), Lossless join and dependency preserving decomposition, Fourth normal form(4NF), Fifth Normal Form (5NF). **[7 Hrs]**

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UNIT V: **(CO5)**

Transaction Management & Recovery: ACID properties, serializability of Transaction, Testing for Serializability and concurrency control, Lock based concurrency control (2PL, Deadlocks), Time stamping methods, Log Based Recovery, Recovery with Concurrent Transactions **[7Hrs]**

Text Books:

S.No.	Title	Author(s)	Publisher
1.	Database System Concepts	Korth & Sudarshan	Tata McGraw Hill
2.	Database Management Systems	Raghurama Krishnan, Johannes Gehrke	Tata McGraw Hill

Reference Books:

S.No.	Title	Author(s)	Publisher
1.	Principles of Database Systems	Ullman, J.O.	Galgotia Publications
2.	Fundamentals of Database Systems	Elmasri R. & Navathe S.B	Pearson Education
3.	[Online] https://onlinecourses.nptel.ac.in/noc22_cs51/preview		

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SYLLABUS

B.Tech. (Mechanical Engineering) Seventh Semester

INNOVATION MANAGEMENT & ENTREPRENEURSHIP					
(Open Elective Course : Open to all branches)					
Course Code	**100747ME**	**L= 3**	**T= 0**	**P= 0**	**Credits= 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3Hours**
	Minimum number of class tests to be conducted=02			**Minimum assignments=02**	

Course Objectives	Course Outcomes
The course will provide a thorough coverage of conceptual framework on Entrepreneurship development and enhance student's innovation skill. This course will help provide a quick understanding of essential concepts and issues. And enhance the students to have an understanding the problems and prospects related to setting up of any type of business.	**On successful completion of the course, the student will be able to:** **CO1:** Explain the basic terms and types of innovation management. **CO2:** Apply the knowledge management Applications. o use in their practices **CO3:** Explain the diverse methods employed in design thinking, establish a workable design thinking framework and demonstrate the critical theories of design, systems thinking, and design methodologies **CO4:** Applying leadership principles and to work in an interdisciplinary environment through different entrepreneurial strategies. **CO5:** Describe of various techniques in entrepreneurship including intellectual property preparation.

UNIT-I : Innovation **CO1**

Introduction to innovation, Concept, need, objectives, Scope of Innovation management- Characteristics, Significance, Principles, Types of Innovation, Factors influencing innovations, Nurturing Innovation in E-business, Sources of Innovation, Determinants, Innovation behavior, Forecasting Tools. **[8Hrs]**

UNIT-II: Knowledge Management **CO2**

An Introduction to Knowledge Management, evolution of Knowledge management, from Information Management to Knowledge Management, key challenges, ethics for Knowledge Management, future of Knowledge Management. **[7Hrs]**

UNIT-III: Design Thinking **CO3**

Basics of Design Thinking, Why Design Thinking, Design Thinking Mindset (Inspiration, Ideation and Implementation) Design thinking process, (Empathy, Define, Ideate, Prototype, Test). Design team-Team formation, Conceptualization- Visual thinking, Drawing/sketching, New concept thinking, Concept Generation Methodologies, Concept Selection, Concept Testing, Opportunity identification Prototyping- Principles of prototyping, Prototyping technologies. **[7Hrs]**

UNIT- IV: Entrepreneurship **CO4**

Who is an entrepreneur? Introduction to entrepreneurship, Role of Entrepreneur, Characteristics of an entrepreneur; Factors affecting entrepreneurial growth – Social, cultural, personality factors, psychological and Social Factors. Impact of entrepreneurship for sustainable development, **Entrepreneurship Development Programme**- Objectives of EDP, Types of EDP, Stages of EDP, Enterprise Launching Formalities-Procedure, Planning your Enterprise, Elements of planning, Business plan in a nutshell, Sources of help for Entrepreneurs. **[7Hrs]**

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UNIT-V: **Problem identification and problem solving** **CO5**

Problem definition, Idea generation using brain storming and mind map, Concept generation using 6-3-5 Brain writing technique and Concept finalization, PESTEL and SWOT analysis, Define the product life cycle by applying the principles of obsolescence management, sustenance engineering and End-of-Life (EoL), Feasibility Report, Considerations while preparing a Feasibility Report, Proforma of Feasibility Report, Role of Institutions and entrepreneurship, Prepare the Intellectual Property Rights (IPR) documents. **[7Hrs]**

Text Books

S.No.	Title	Author(s)	Publisher
1.	Innovation Management	Dr.C.S.G. Krishnacharyulu & Dr. R. Lalitha	Himalaya Publishing House
2.	Entrepreneurship Management	Dr. Aruna Kaulgud	Vikas Publishing House

ReferenceBooks:

S.No	Title	Author(s)	Publisher
1.	Design Thinking Methodology Book	EmrahYayici	Atlantic Publishers and Distributors
2.	Knowledge Management for the Information Professional	Srikantaiah, T.K., Koenig, M	Information Today
3.	Change by Design: How Design Thinking Transforms Organizations and Inspires Innovation	Tim Brown	HarperCollins Publishers
4.	Entrepreneurship & Innovation Management – An Industry Perspective	Gopal.R, Pradip Manjrekar	Excel Books
5.	Entrepreneurship Management and Development	Ajith Kumar	Himalaya Publishing House
6.	Innovation and Entrepreneurship Practice and Principles	Drucker, Petere	East West Press
7.	Managing technology innovation	Ettlite I E	John Willey & Sons

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SYLLABUS

B.Tech. (Seventh Semester)

ANDROID APPLICATION DEVELOPMENT					
(Open Elective Course : Open to all branches except CSE & IT)					
Course Code	**100748CA**	**L = 3**	**T = 0**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
The objective of this course is to make the student capable of programming in Android, using the required features of today's smart phones.	**On successful completion of the course, the student will be able to:** **CO1:** Identify the basic architecture and installation process **CO2:** Explain the various user interface **CO3:** Identify the different dialog controls and other features **CO4:** Apply the concept of data storage. **CO5:** Describe the implementation of networking and google play service.

UNIT-I **(CO1)**

Android Introduction, Smartphones future, Android Architecture, Preparing the Environment, Installing Android Studio, Installing the SDK, Creating Android Emulator, Activity life cycle, Creating a project, Working with the AndroidManifest.xml. **[8 Hrs]**

UNIT-II **(CO2)**

Introduction to UI- Layouts, Fragments, Adapters, Action bar, Dialogs, Notifications, Supporting multiple screen sizes. Designing User Interface Using Views- TextView, Button, Image Button, Check Box, Toggle Button, Radio Button etc., Progress Bar View and Auto Complete Text View, Time Picker and Date Picker View, List View, Image View, Intents, Status bar notifications, Toast notifications. **[7 Hrs]**

UNIT-III **(CO3)**

Menus, Localization, Options menu, Context menu Dialogs- Alert dialog, Custom dialog, Dialog as Activity Orientation and Movement- Pitch, roll and yaw, Natural device orientation, SMS - Sending and Receiving, Working with Media– Playing audio and video. **[7 Hrs]**

UNIT-IV **(CO4)**

Services- Service lifecycle, Foreground service, Location and Maps - Google maps, Using GPS to find current location, Working with data storage, Using External storage, SQLite database, Files access, Animation, Working with Sensors, Working with Camera. **[7 Hrs]**

UNIT-V **(CO5)**

Content providers- Content provider introduction, Query providers, Network Communication- Web Services, HTTP Client, XML and JSON, Creating own services, Publishing and Distributing Your App- Preparing for publishing, Google Play requirements, Signing and preparing the graphics, Publishing to the Android Market, Monetization, Tips on becoming a top app, Google analytics. **[7 Hrs]**

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Chairman (AC)	Chairman (BoS)	Date of Release	Version	AY 2023-24 Onwards

Text Books:

S.No.	Title	Author(s)	Publisher
1.	Learning Android	Ramesh Bangia	Khanna Publisher
2.	Programming Android	Zigurd Mednieks, Laird Dornin, G. Blake Meike, Masumi Nakamura	O'Reilly

Reference Books:

S.No.	Title	Author(s)	Publisher
1.	Android App Development For Dummies	Michael Burton	Wiley
2.	Headfirst Android Development	Dawn Griffiths	O′Reilly

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Bhilai Institute of Technology, Durg

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SYLLABUS

B. Tech. (Information Technology) Seventh Semester

CRYPTOGRAPHY & NETWORK SECURITY LAB					
Course Code	**106791IT**	**L = 0**	**T = 0**	**P = 2**	**Credits = 1**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	25	**-**	**25**	**50**	**-**

Course Objectives	Course Outcomes
To make the student learn different encryption techniques along with hash functions, MAC, digital signatures and their use in various protocols for network security and system security	**On successful completion of the course, the student will be able to:** **CO1:** Provide security of the data over the network. **CO2:** Identify basic security attacks and services. **CO3:** Use symmetric and asymmetric key algorithms for cryptography **CO4:** Build cryptosystems by applying symmetric and public key encryption algorithms. **CO5:** Construct code for authentication algorithms.

List of Experiments

(At least Ten experiments are to be performed by each student)

1.	Write a program in C/C++ to input a String(array of character) and perform XOR operation on each character of string with 0.	**(CO1)**
2.	Write a program in C/C++ to input a String(array of character) and perform XOR operation on each character of string with 127.	**(CO1)**
3.	Write a program in C/C++ to input a String(array of character) and perform AND operation on each character of string with 127.	**(CO2)**
4.	Write a program in C/C++ to implement Euclid algorithm.	**(CO2)**
5.	Write a program in C/C++ to implement Extended Euclid algorithm.	**(CO3)**
6.	Write a program in C/C++ to implement substitution cipher (Polyalbhabatic) technique.	**(CO3)**
7.	Write a program in C/C++ to implement conventional cipher (Caesar cipher) technique.	**(CO3)**
8.	Write a program in C/C++ to implement transposition cipher (Columnar) technique.	**(CO3)**
9.	Write a program in C/C++ to implement Chinese Remainder Algorithm.	**(CO3)**

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10.	Write a program in C/C++ to implement RSA Algorithm.	**(CO4)**
11.	Write a program in JAVA to implement RC4 Ecncyption Technique.	**(CO4)**
12.	Write a program in JAVA to detect the MAC address of any system on the network.	**(CO4)**
13.	Write a program in C/C++ to implement Play Fair Cipher Technique.	**(CO5)**
14.	Write a program in C/C++ to implement Diffie-Hellman key Exchange Algorithm.	**(CO5)**
15.	Write a program in C/C++ to check whether the numbers are relatively prime/coprime or not.	**(CO5)**

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SYLLABUS

B. Tech. (Information Technology) Seventh Semester

CLOUD COMPUTING LAB					
Course Code	**106792IT**	**L = 0**	**T = 0**	**P = 2**	**Credits = 1**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	25	**-**	**25**	**50**	**-**

Course Objectives	Course Outcomes
Identify the technical foundations of cloud systems architectures, cloud application problems, application design and management and technical challenges for cloud applications and assess their importance.	**On successful completion of the course, the student can**: **CO1.** Configure various virtualization tools such as Virtual Box, VMware workstation. **CO2.** Design and deploy a web application in a PaaS environment. **CO3.** Simulate a cloud environment to implement new schedulers. **CO4.** Install and use a generic cloud environment that can be used as a private cloud. **CO5.** Manipulate large data sets in a parallel environment.

List of experiments: -

1. Install Virtualbox/VMware Workstation with different flavours of linux or windows OS on top of windows. **[CO1]**
2. Install a C compiler in the virtual machine created using virtual box and execute Simple Programs. **[CO1]**
3. Install Google App Engine. Create hello world app and other simple web applications using python/java. **[CO2]**
4. Use GAE launcher to launch the web applications. **[CO2]**
5. Simulate a cloud scenario using CloudSim and run a scheduling algorithm that is not present in CloudSim. **[CO3]**
6. Find procedure to run the virtual machine of different configuration. Check how many virtual machines can be utilized at particular time. **[CO5]**
7. Find procedure to attach virtual block to the virtual machine and check whether it holds the data even after the release of the virtual machine. **[CO5]**
8. Install a C compiler in the virtual machine and execute a sample program. **[CO1]**
9. Find procedure to install storage controller and interact with it. **[CO3]**
10. Find a procedure to launch virtual machine using trystack (Online Openstack DemoVersion) **[CO1]**
11. Install Hadoop single node cluster and run simple applications like wordcount. **[CO4]**

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SYLLABUS

B. Tech. (Information Technology) Seventh Semester

CAPSTONE PROJECT PHASE - 1					
Course Code	**106793IT**	**L = 0**	**T = 0**	**P = 4**	**Credits = 2**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	50	**-**	**50**	**100**	**-**

Course Objectives	Course Outcomes
The Course objective is to provide an opportunity for students to understand real-life practices and implementation of theoretical Engineering knowledge in practical environment. To provide an opportunity for students to acquire interpersonal skills and ability for team work. To obtain knowledge of how to make optimal decisions to resolve technical challenges.	**On successful completion of the course, the student will be able to:** **CO1:** Apply engineering knowledge in real life projects by integrating theoretical concepts with real-time situations. **CO2:** Acquire practical skills and Communication skills while working on projects. **CO3:** Prepare and Produce decisions to resolve technical challenges. **CO4:** Write technical documents and give oral presentations related to the project work completed. **CO5**: Use of Advanced Tools for developing better projects.

● Minor project is introduced in the fifth semester with a specific objective to strengthen the understanding of student's fundamentals through effective application of theoretical concepts of Programming Language.

● Student Groups with 3 or 4 members should identify a topic of interest in consultation with Faculty Guide.

● Review the literature and gather information pertaining to the chosen topic.

● State the objectives and develop a methodology to achieve the objectives.

● Carryout the design/ develop codes/programs to achieve the objectives.

● Demonstrate the novelty of the project through the results and outputs.

● The progress of the project is evaluated based on a minimum of two reviews.

● The review committee may be constituted by the Head of the Department.

● A project report is required at the end of the semester.

● The product has to be demonstrated for its full design specifications.

● The internal evaluation will be made based on the product, the report and a viva-voce examination, conducted internally by a committee appointed by Head of the Department comprising HoD or a Project coordinator for that program.

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SYLLABUS

B. Tech. (Information Technology) Seventh Semester

INTERNSHIP ASSESSMENT (REPORT & SEMINAR)					
Course Code	**106794IT**	**L =0**	**T =0**	**P =2**	**Credits =1**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	-	-	25	25	-

Course Objectives	Course Outcomes
The course objective is to promote cooperation and to develop synergetic collaboration between industry and the university in promoting a knowledgeable society. • To expose students to the 'real' working environment and get acquainted with the organization structure, business operations and administrative functions. • To have hands-on experience in the students' related field so that they can relate and reinforce what has been taught at the college and • To obtain knowledge of how to make optimal decisions to resolve work challenges.	**On successful completion of the course, the student will be able to:** **CO1**: Apply engineering knowledge in solving real-life problems. **CO2**: Attain new skills and be aware of the state-of-art in engineering disciplines of their own interest. **CO3**: Describe use of advanced tools and techniques encountered during training and visit. **CO4**: Acquire practical skills, organizational skills, Communication skills, lifelong learning skills, professional awareness and experience working on projects and alongside industry experts. **CO5:** Write a technical project report that follows an established structure and give oral presentations with focus on the project results and a credible work procedure.

1. As per institution's guidelines, Industrial Training have to be done at the end of 6th Semester.
2. The student must follow the instructions given by the Teacher In-charge of Industrial Training.
3. The purpose of the Industrial Training is to develop the work process being performed and apprise them of the industry problems.
4. During the training, students will be given practical problems by the industry in which they are undergoing training. In case the industry do not give them the problems, the students will themselves formulate problems and carry out detailed study on them and recommend the optimum solution based on their theory knowledge.
5. Submission of Final Report - Each student must submit a Final Report and deliver a presentation on industrial training at the end of the industrial training period on the date and time announced by the Teacher In-charge.
6. The Project report must include the following:
   (a) The basic history/introduction of the industry.
   (b) The software and hardware used.
   (c) The sequence of operations followed/ systems introduced for the project development.
   (d) The formulation of practical problems.
   (e) Data required formulating the problems.
   (f) Analysis of the data, steps required and commands used in industry.
   (g) Certificate from the industry for the period of training undergone.

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The student would be evaluated through Report and Viva-voce.

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SYLLABUS

B. Tech. (Common to all branches) Seventh Semester

UNIVERSAL HUMAN VALUES & PROFESSIONAL ETHICS					
Course Code	**100795HM**	**L = 0**	**T = 0**	**P = 2**	**Credits = -**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	-	**-**	**25**	**25**	**-**

Course Objectives	Course Outcomes
. To introduce appropriate inputs of value education to students for developing the right understanding about oneself and the rest of reality through self-exploration, realization of inherent coexistence, harmony and self-regulation at various levels to enable them to achieve their cherished goals and live a fulfilling life.	**On successful completion of the course, the student will be able to:** **CO1:** Demonstrate understanding about the need, basic guidelines, content and process of value education. **CO2:** Distinguish between the Self and the Body, and understand its harmony and co-existence. **CO3:** Demonstrate the understanding about the value of harmonious relationship to explore their role in ensuring a harmonious society. **CO4:** Demonstrate understanding about harmony in nature and existence and their mutually fulfilling participation in nature. **CO5:** Distinguish between ethical and unethical practices, to actualize a harmonious environment at their functional areas

UNIT – I Introduction – Need and Basic Guidelines **CO1**

- Value Education: Need, Basic Guidelines, Content and Process
- Self-Exploration: Definition, Content and Process
- Natural Acceptance and Experiential Validation: Mechanism for Self Exploration
- Continuous Happiness and Prosperity: Basic Human Aspirations, Right understanding about Relationship and Physical Facilities- the basic requirement for fulfillment of aspirations of every human being with their correct priority. **[5Hrs]**

UNIT – II Harmony in Human Being **CO2**

- Human Being: Co-existence of Self (I) and the material 'Body', Understanding the needs of Self and Body-Sukh and Suvidha, Body as an instrument of 'I'.
- Characteristics and activities of 'I': Harmony within oneself and with the Body (Sanyam and Swasthya), Programs to attain 'Sanyam and Swasthya'.
- Appraisal of Physical Needs: Meaning of prosperity in detail. **[5Hrs]**

UNIT – III Harmony in Relationships – Family and Society **CO3**

- Harmony in family: Basic unit of human interaction; meaning of Nyaya (Righteousness); program for its fulfillment to ensure Ubhay Tripthi (Fulfillment of both material and spiritual aspirations)
- Harmony in Society: Understanding Society as extension of family, Samadhan (Solution), Samridhi (Prosperity), Abhay (Fearlessness), Sah-astitva (Co-existence) as comprehensive Human Goals
- Trust and Respect: Foundational values of relationship; Understanding the meaning of Vishwas (Trust) and Samman (Respect); other salient features in relationship.
- Universal Harmonious order in Society: Akhand Samaj (Undivided Society), Sarvabhaum Vyawastha (Universal Order) – from family to world. **[5Hrs]**

UNIT – IV Harmony in Nature and Existence **CO4**

- Harmony in Nature: Understanding the harmony in nature; interconnectedness and mutual fulfillment
- Interconnection among the Four Orders of Nature: Mutual fulfillment of the four order of nature – Material Order (Padartha avastha), Plant Order (Pranic avastha), Animal Order (Jeeva avastha) and Human Order (Gyan avastha); Recycling and Self-regulation in Nature
- Existence as Co-existence(Sahastitva): Existence of mutually interacting units, Holistic perception of harmony at all levels of existence. **[5Hrs]**

UNIT – V Professional Ethics and Implications of Harmony on it **CO5**

- Definiteness of Ethical Human Conduct: Natural acceptance of human values; basis for Humanistic Education, Humanistic Constitution and Humanistic Universal Order
- Competence in Professional Ethics: Professional competence for augmenting universal human order, people friendly and eco-friendly production systems, technologies and management models; Case studies of typical holistic technologies, management models and production systems
- Strategy for transition from the present state to Universal Human Order: At the individual level (as socially and ecologically responsible engineers, technocrats and managers; at the societal level (as mutually enriching institutions and organizations) **[4Hrs]**

Text Books:

S. No.	Title	Author(s)	Publisher
1.	A foundation Course in Human Values and Professional Ethics	R R Gaur, R Sangal & G P Bagaria	Excel Books

Reference Books:

S. No.	Title	Author(s)	Publisher
1.	Jeevan Vidya Ek Parichay	A Nagaraj	Divya Path Sansthan
2.	Fundamentals of Ethics for Scientists & Engineers	E G Seebauer & Robert L Berry	Oxford University Press
3.	Science and Humanism	P L Dhar & R R Gaur	Commonwealth Publishers
4.	Engineering Ethics (including Human Values)	M Govindrajan, S Natarajan & V S Senthil Kumar	Prentice Hall of India
5.	Human Values	A N Tripathy	New Age International
6.	Indian Ethos and Modern Management	B L Bajpai	New Royal Book Co.